

Dharmanshu Kumar

B.Tech CSE Student · Distributed Systems & Networking · Full-Stack

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SUMMARY

Computer Science undergraduate at SRMIST focused on **distributed systems, database internals, and low-level networking**. Engineered a Raft consensus key-value store with formal TLA+ specifications and WAL crash-recovery, zero-dependency DPI-resistant VPN tunnels in Rust and Go with mTLS and SNI camouflage, and out-of-core data linkage pipelines under strict memory bounds. Experienced in on-device acoustic ML on Android and operating self-hosted edge infrastructure on Oracle Cloud.

EDUCATION

Bachelor of Technology (B.Tech) - Computer Science & Engineering 2025 - 2029
SRM Institute of Science and Technology (SRMIST)

Relevant Coursework & Technical Focus Areas:

- Distributed Computing
- Computer Networks
- Operating Systems
- Database Management Systems
- Network Protocols and Algorithms
- Data Structures and Algorithms
- Computer Organization and Architecture

TECHNICAL SKILLS

DATABASE & DATA ENGINEERING

- SQL & Relational Schema Design
- Write-Ahead Logging & Crash Recovery
- Out-of-Core Pipelines (DuckDB / Polars / Parquet)
- Record Linkage & Inverted Index Blocking

LANGUAGES

- Go: Concurrency & Protocol Design
- Rust: Systems Programming & Memory Safety
- Python: Data Pipelines & Automation
- TypeScript / JavaScript: React & Node.js
- C / C++ / Java: Coursework & DSA

NETWORKING & SYSTEMS

- Distributed Consensus (Raft)
- Custom TCP/UDP VPN Protocol Design
- TLS / mTLS Mutual Auth & Traffic Camouflage
- Low-Level Sockets & WebSocket Telemetry

WEB & INFRASTRUCTURE

- React.js & Node.js / Express Architectures
- Oracle Cloud (OCI VPS), Nginx, systemd
- Android SDK (Jetpack Compose) & TensorFlow Lite
- Microsoft 365 E5 & Azure API Automation

FEATURED ENGINEERING PROJECTS

GO
2026

raft-kv

Implemented the Raft distributed consensus algorithm from scratch in Go without consensus libraries: leader election, log replication, snapshot installation, and linearizable read log proposals with client request deduplication. Engineered a CRC32-framed write-ahead log (WAL) that detects and recovers from crash-torn writes. Validated protocol invariants using a formal **TLA+ model (31,645 states checked, 0 violations)** and an automated chaos test harness injecting network partitions and process kills.

- Go
- Raft Consensus
- TLA+
- Write-Ahead Log
- Chaos Testing
- Linearizability

FEATURED ENGINEERING PROJECTS (CONTINUED)

RUST / GO
2025 - 2026

shadowlink / tunnel

Architected a zero-dependency, DPI-resistant VPN tunnelling system implemented in both Rust and Go. The Rust engine (shadowlink) utilizes ChaCha20-Poly1305 encryption, TLS SNI camouflage, Wintun integration, and SOCKS5 UDP routing. The Go implementation (tunnel) features mutual TLS with certificate pinning, Yamux stream multiplexing, and compiled Android modules via gomobile. Maintained latency overheads of **20-50 ms on domestic Indian nodes** and **~230 ms to European OCI endpoints** in Sweden.

Rust Go mTLS SNI Camouflage Yamux gomobile OCI VPS

PYTHON
2026

Entity Resolution Pipeline (Amazon ML Challenge 2026)

Built an out-of-core record linkage and deduplication pipeline resolving 1M+ noisy entity listings under strict memory bounds (<4 GB RAM). Designed a two-tier blocking scheme pairing tokenized inverted indexes with phonetic hashing, pruning comparison complexity by over 99.9%. Engineered stream-processing workflows using DuckDB, Polars, and Parquet to extract IDF-weighted n-gram overlaps and string similarity features, feeding candidate pairs into gradient-boosted decision trees.

Python DuckDB Polars Parquet Inverted Index Record Linkage LightGBM

KOTLIN
2026

NoiseStash

Developed an on-device acoustic metering and environmental sound classification Android app with TensorFlow Lite (YAMNet), ensuring zero audio data leaves the phone. Implemented a dual-rate processing loop with real-time A-weighting decibel filters, an OSHA/NIOSH cumulative sound-exposure engine, background microphone foreground services, and local Room persistence.

Kotlin Android SDK Jetpack Compose TensorFlow Lite YAMNet Room

JAVASCRIPT
2026

NetPulse

Built a self-hosted network telemetry and WAN benchmarking platform. Streams non-compressible random memory buffers directly to TCP sockets to neutralize transparent ISP caching and proxy compression. Operates a 50 Hz WebSocket ping loop measuring round-trip jitter standard deviation and packet loss. Deployed on an Oracle VPS with Nginx proxy buffering disabled and systemd management, coupled with a Cloudflare Pages frontend.

Node.js WebSockets Express Nginx systemd Cloudflare Pages

TYPESCRIPT
2026

StockFlow AI / Mobile App (Team of 6 · Hackathon)

Collaborated in a team of 6 on a disaster-resilient supply chain system. Built the mobile driver application with React 19, TypeScript, and Vite, featuring offline route tracking with Leaflet interactive maps and Supabase synchronization. Engineered the backend hazard intelligence service using FastAPI, combining flood and landslide heuristic risk scoring with Google Gemini API analysis to calculate dynamic safe re-routing.

React TypeScript FastAPI Gemini API Leaflet Supabase

ADDITIONAL PROJECTS

- HyperShare** (Node.js, Express): Browser-to-PC local file streaming over Wi-Fi 6, sustaining 2-87 MB/s without cloud relays; packaged as a standalone Windows executable via pkg.
- urlshrtnr** (Node.js): Self-hosted URL shortener deployed on an Oracle VPS behind Nginx with TLS, using atomic-rename file persistence and a Cloudflare Workers edge redirect tier.
- Property Operations Platform** (React, Firestore): Client-facing tenant, room, and payment management dashboard featuring real-time document synchronization and integer-cent financial ledger tracking.
- OCEWS & TurtleBot 4 Pursuit-Evasion** (React, ROS 2): WebGL orbital-collision simulator based on Keplerian astrodynamics and NASA breakup models (Three.js), and a differential-drive robot pursuit controller in ROS 2 with EKF state estimation.